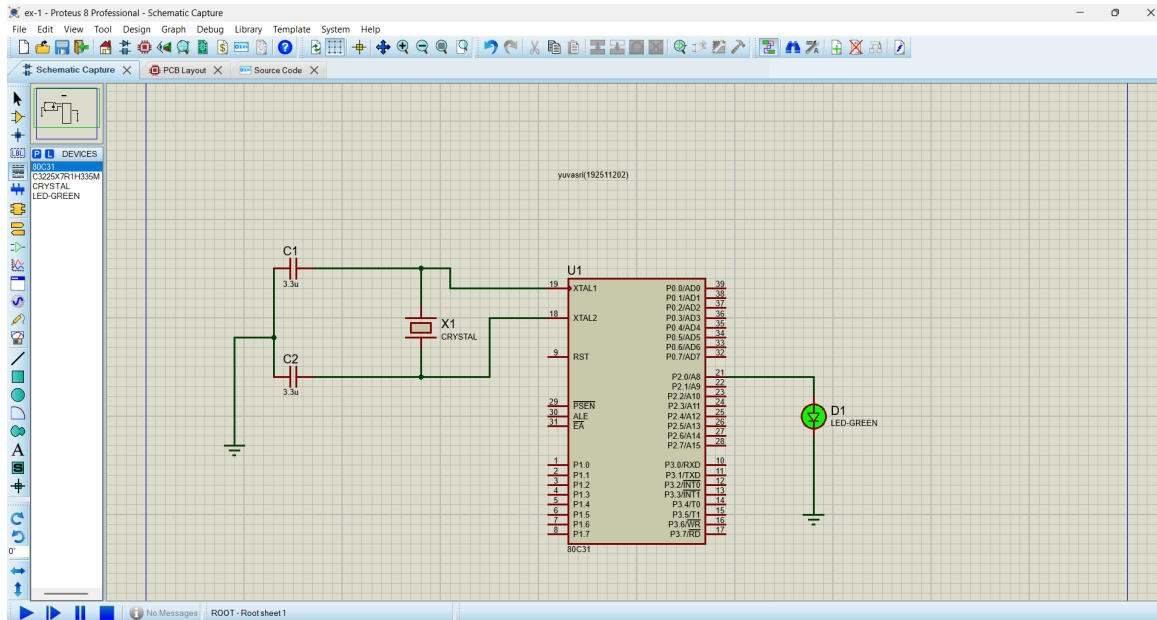


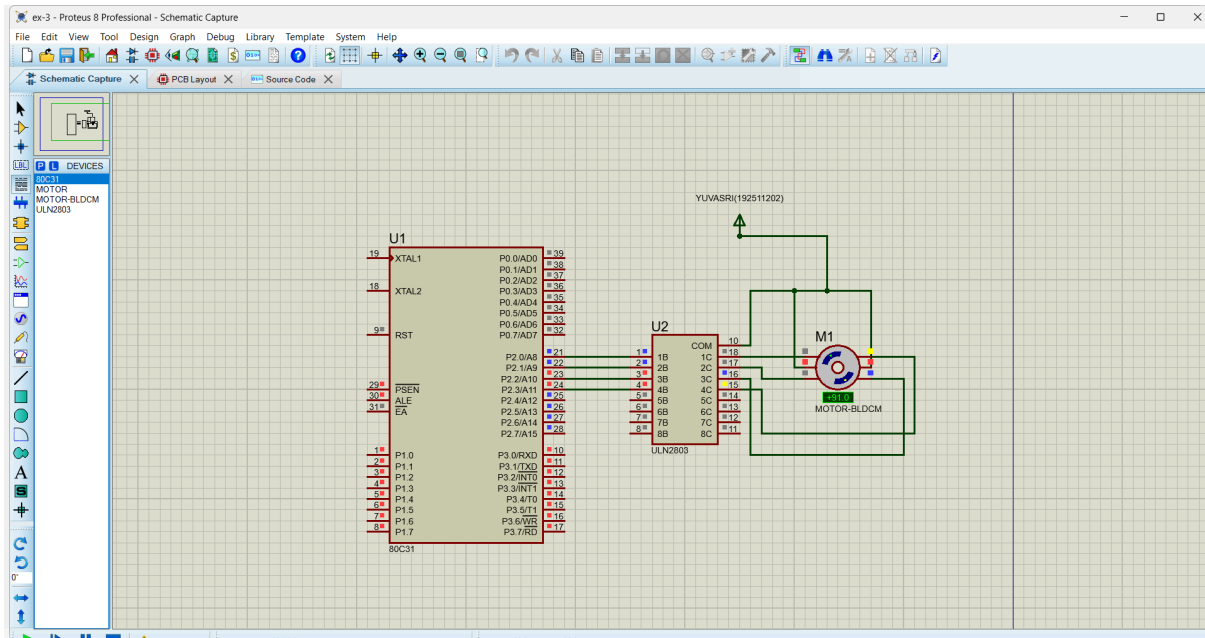
ECA14 – EMBEDDED SYSTEMS

List of Experiments

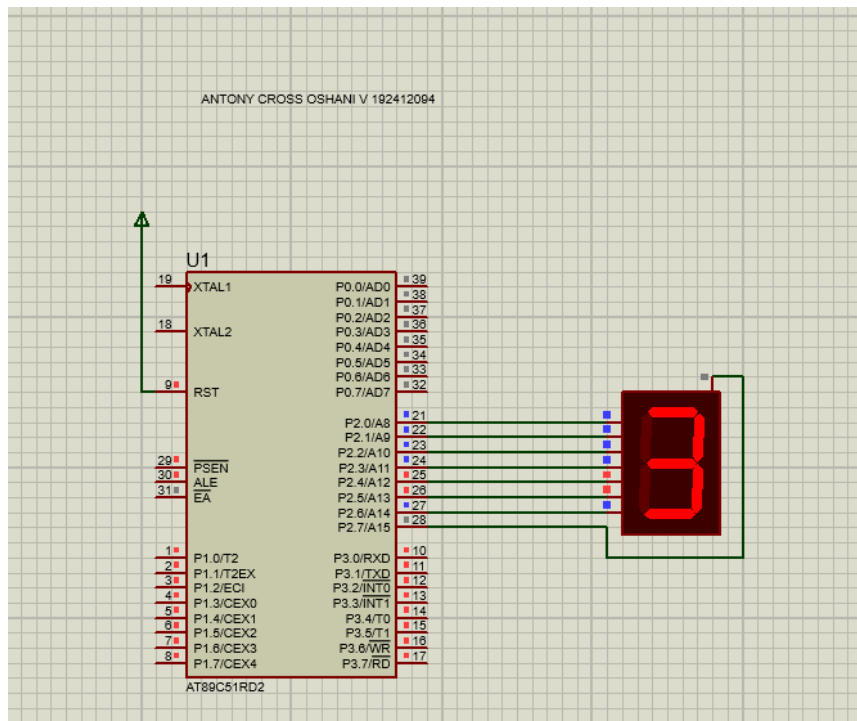
EX-1. 1 Flashing of Led using AT89C51 Microcontroller using Proteus



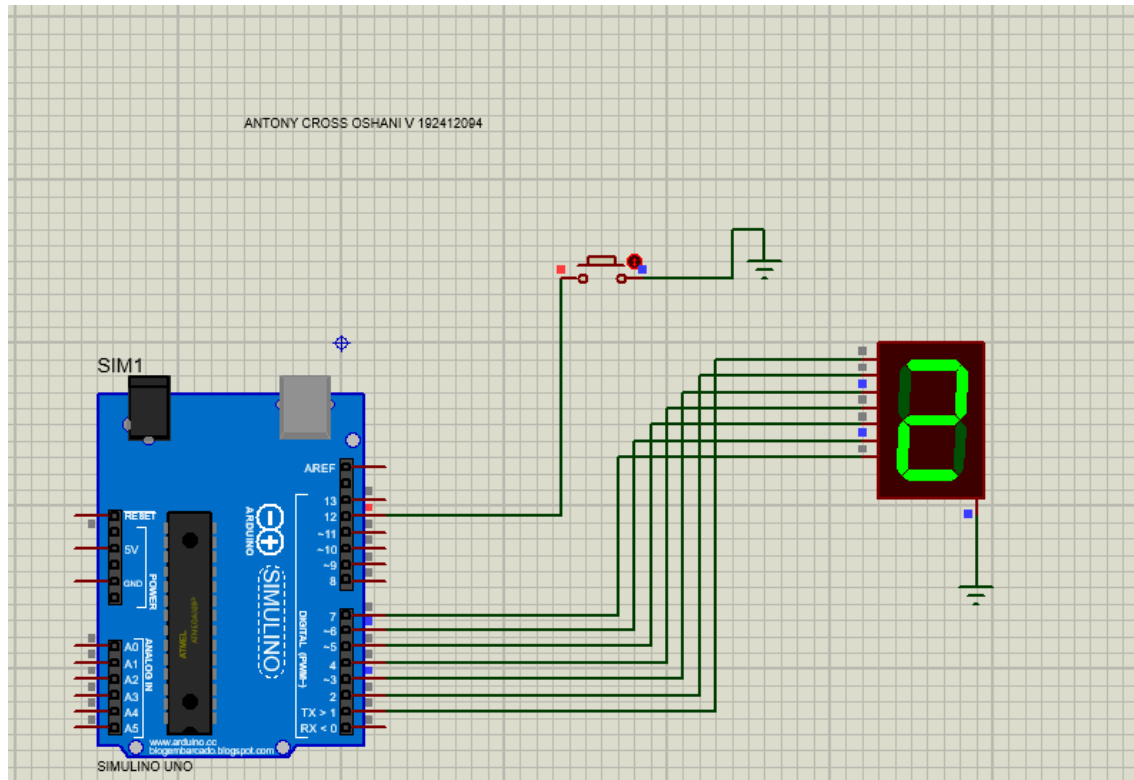
EXP 3: STEPPER MOTOR USING AT89C51 USING PROTEUS



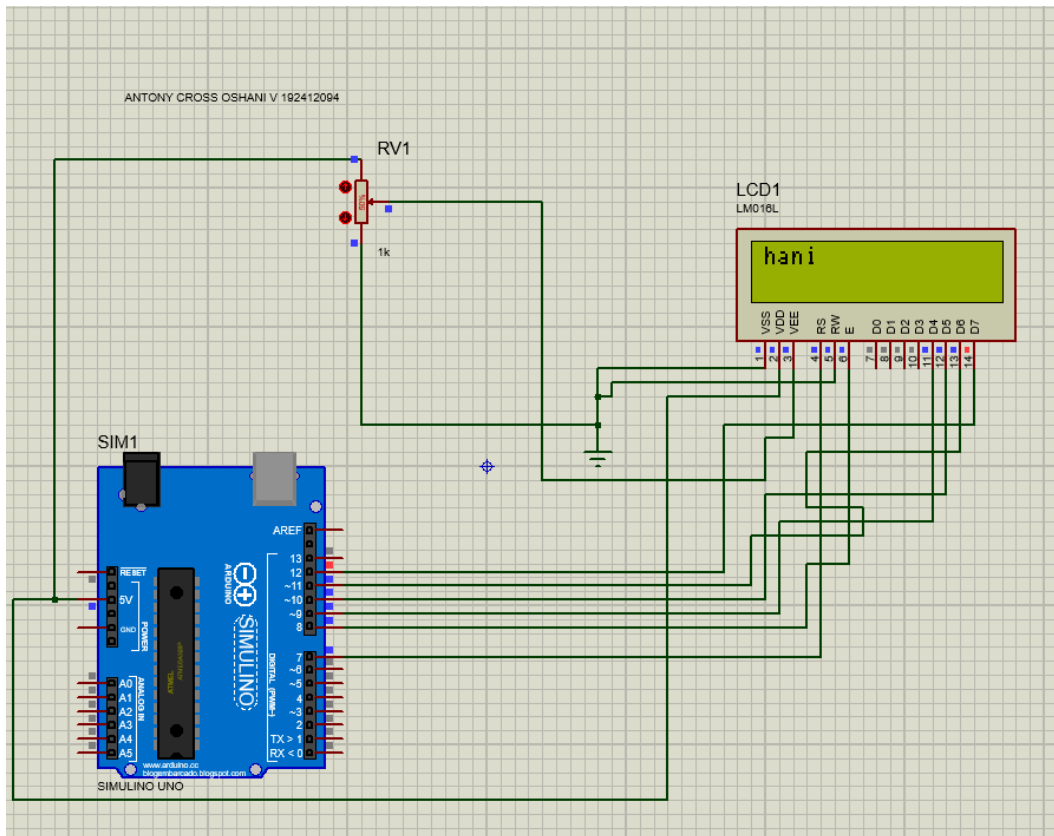
EXP 4: 7 SEGMENT DISPLAY USING AT89C51 USING PROTEUS



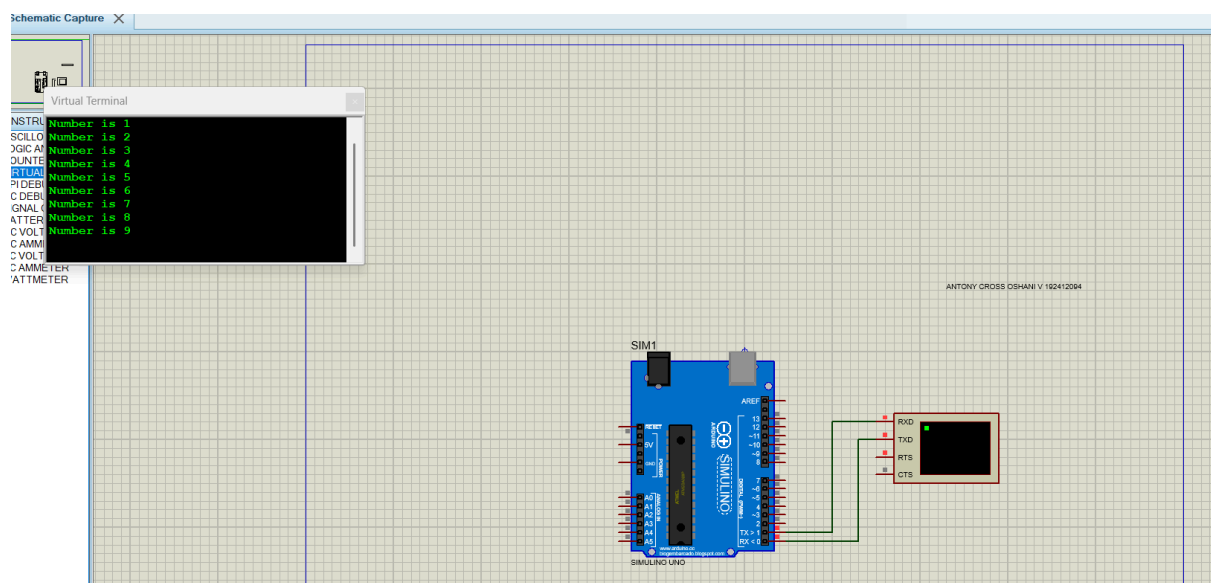
EXP 5: ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS



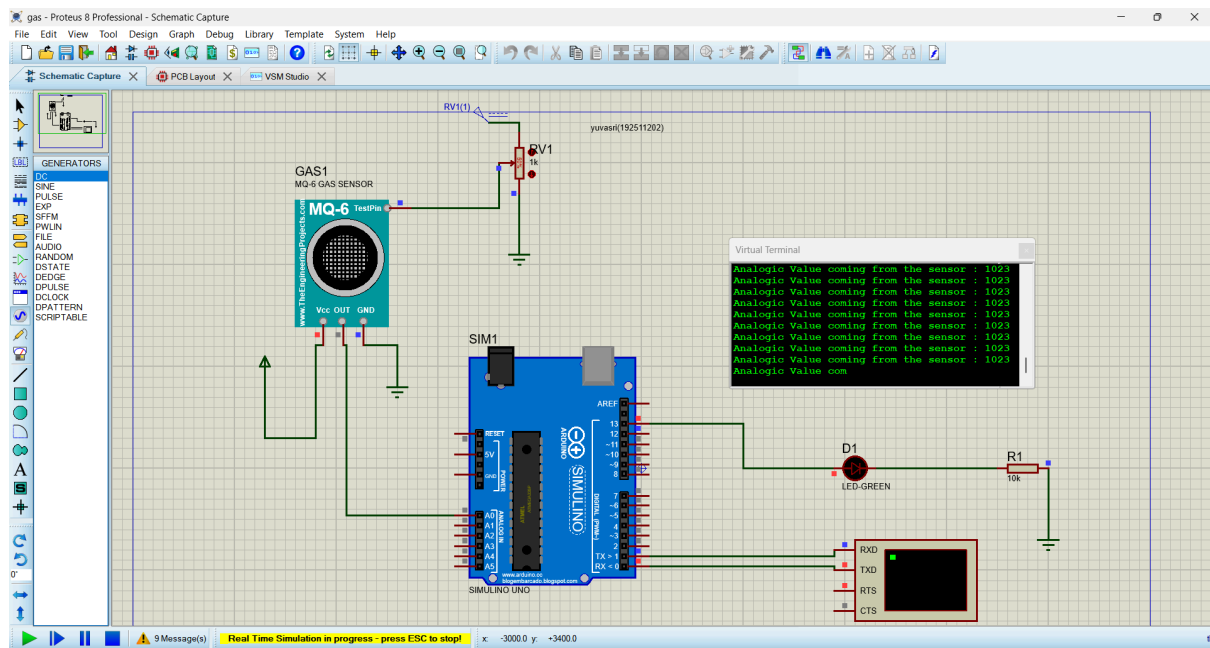
EXP 6 : ARDUINO TO 16x2 LCD DISPLAY USING PROTEUS



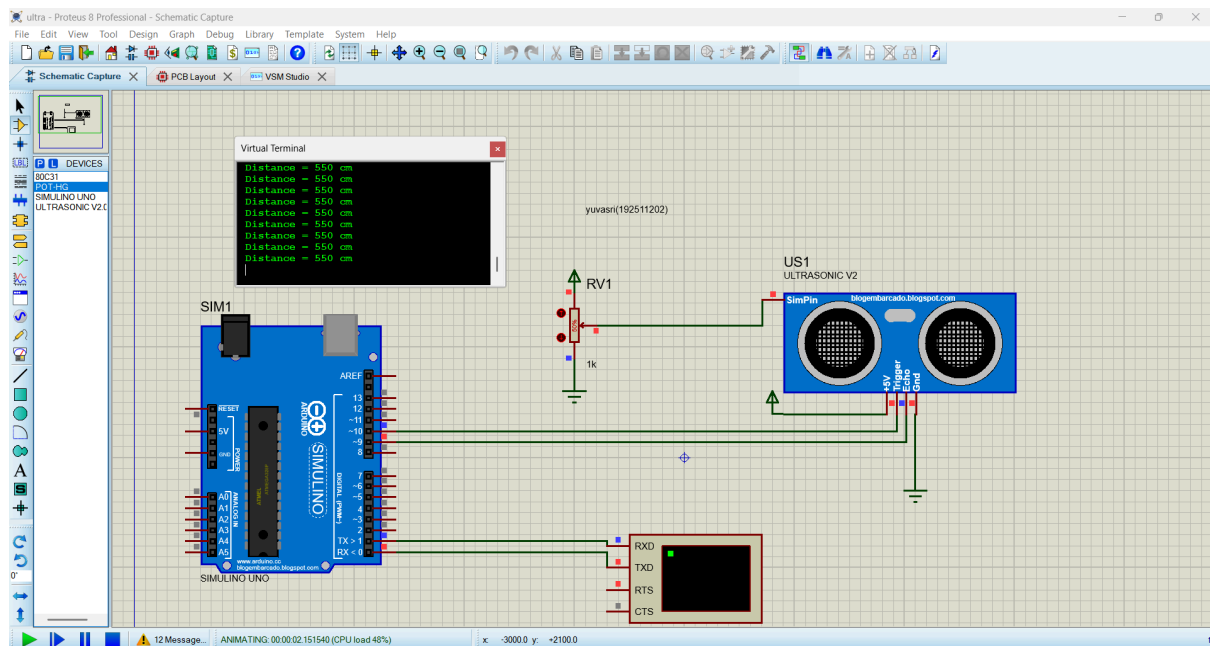
EXP 7: SERIAL COMMUNICATION USING ARDUINO WITH PROTEUS



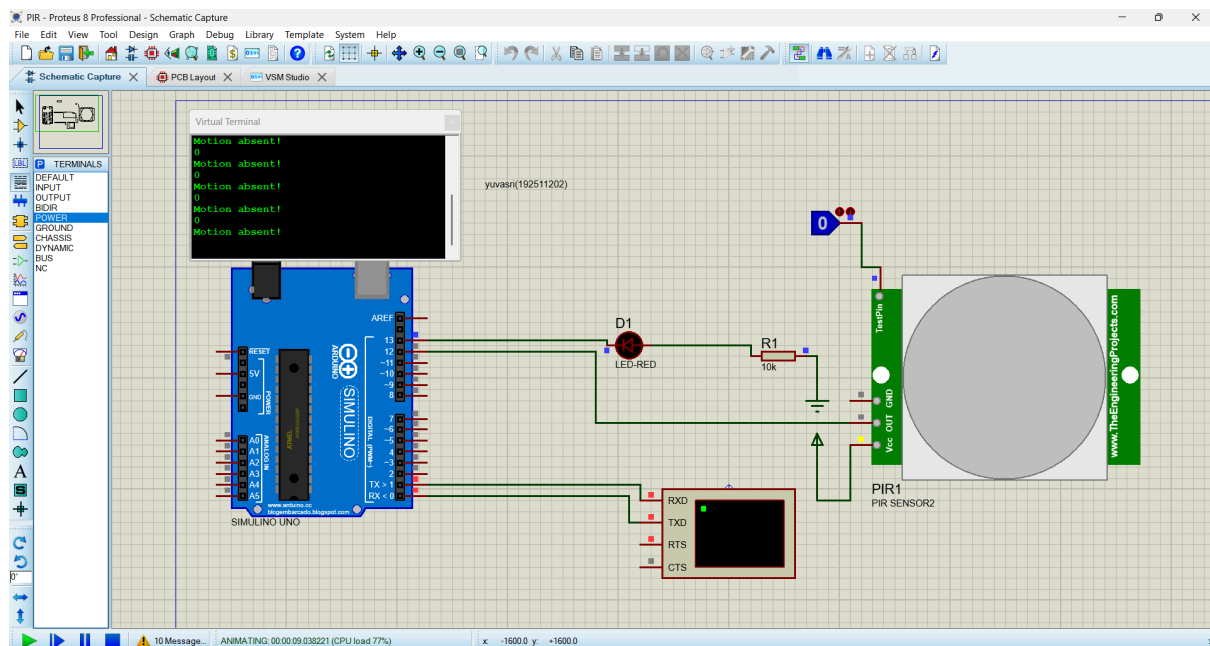
EXP 8 : GAS SENSOR MQ-2 IN PROTEUS USING ARDUINO



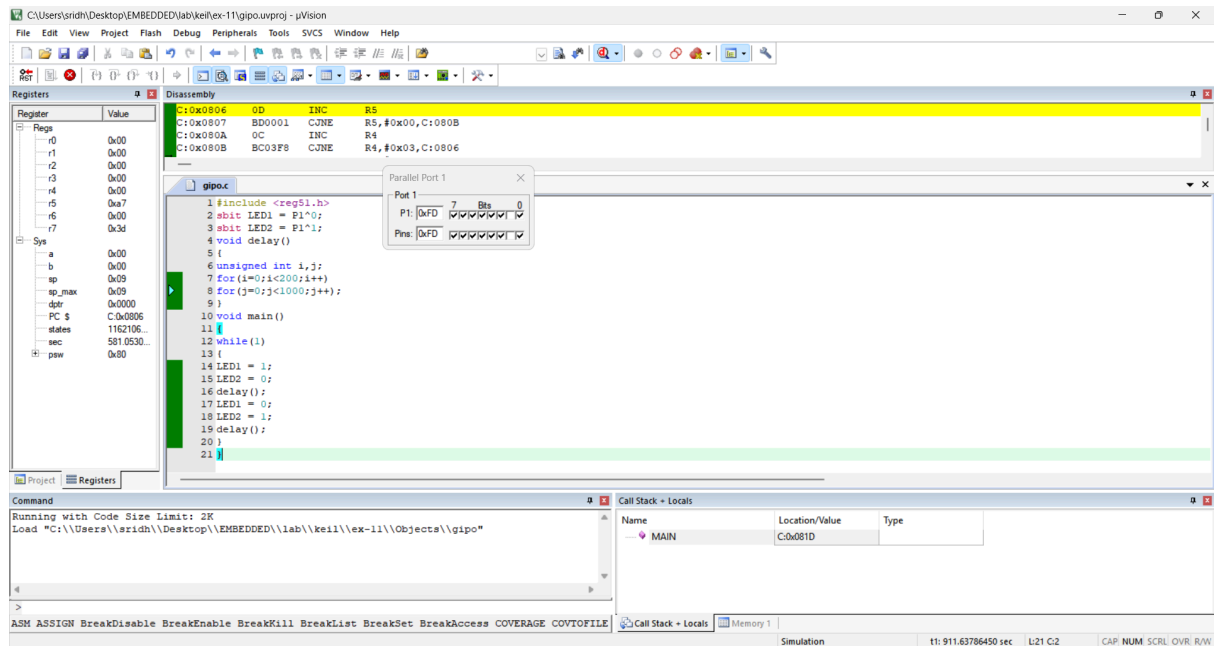
EXP 9 : ULTRASONIC SENSOR IN PROTEUS USING ARDUINO



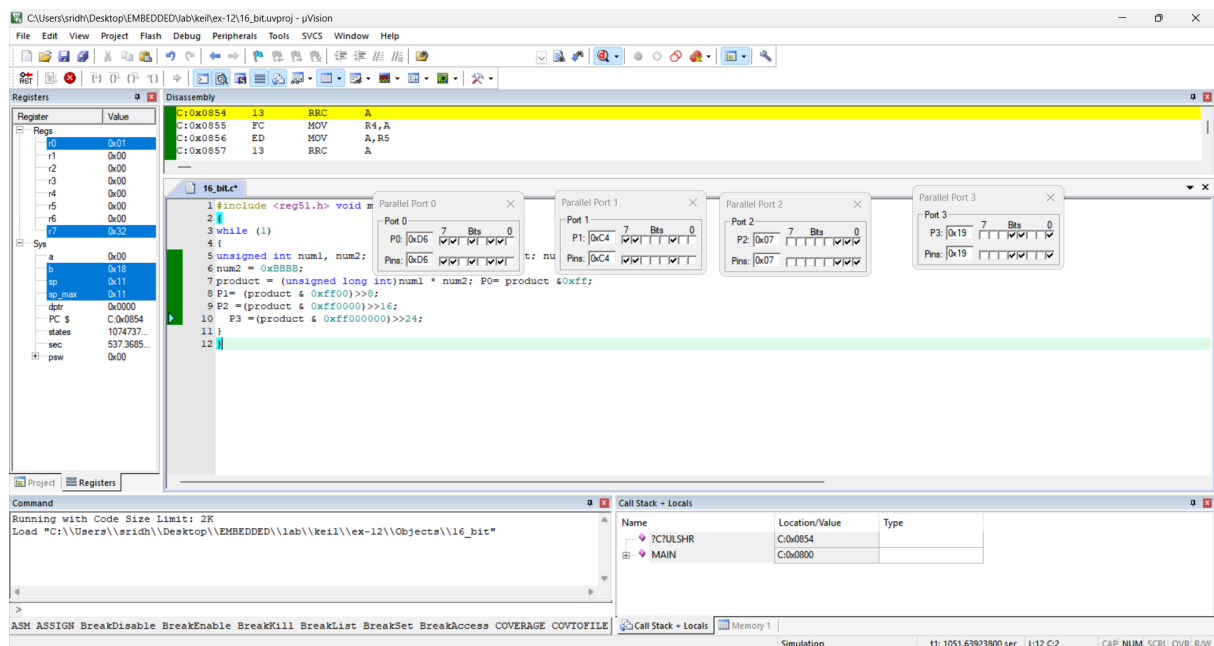
EXP 10 : PIR SENSOR IN PROTEUS USING ARDUINO



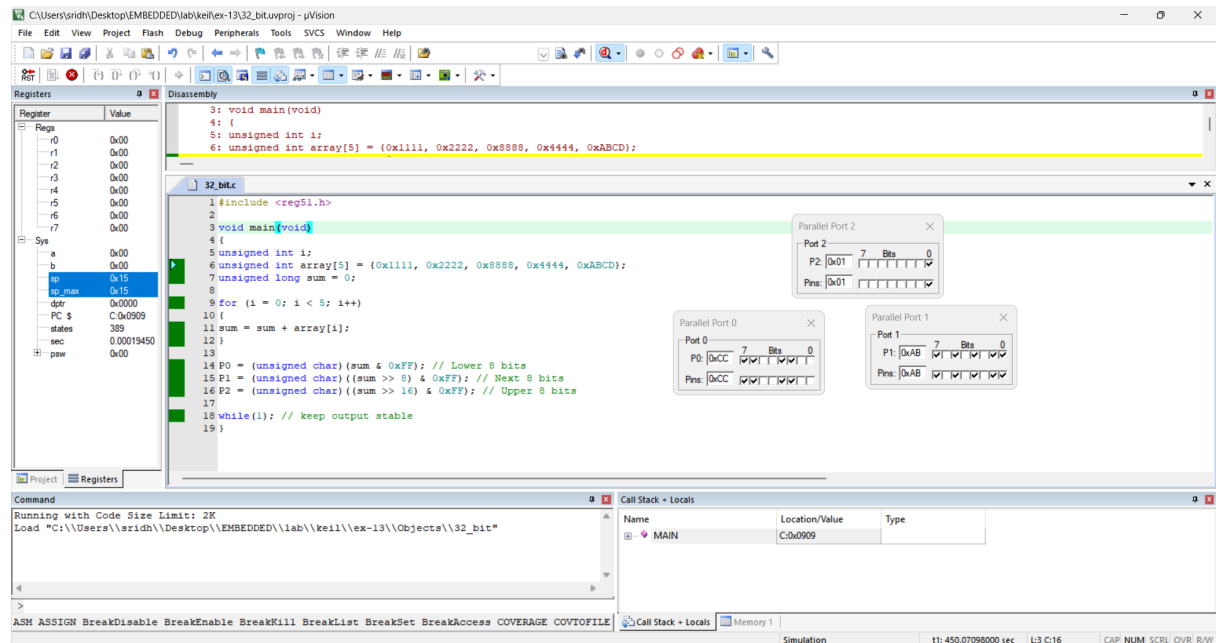
EXP 11 : STUDY OF GPIO & BIT ADDRESSING IN AT89C51



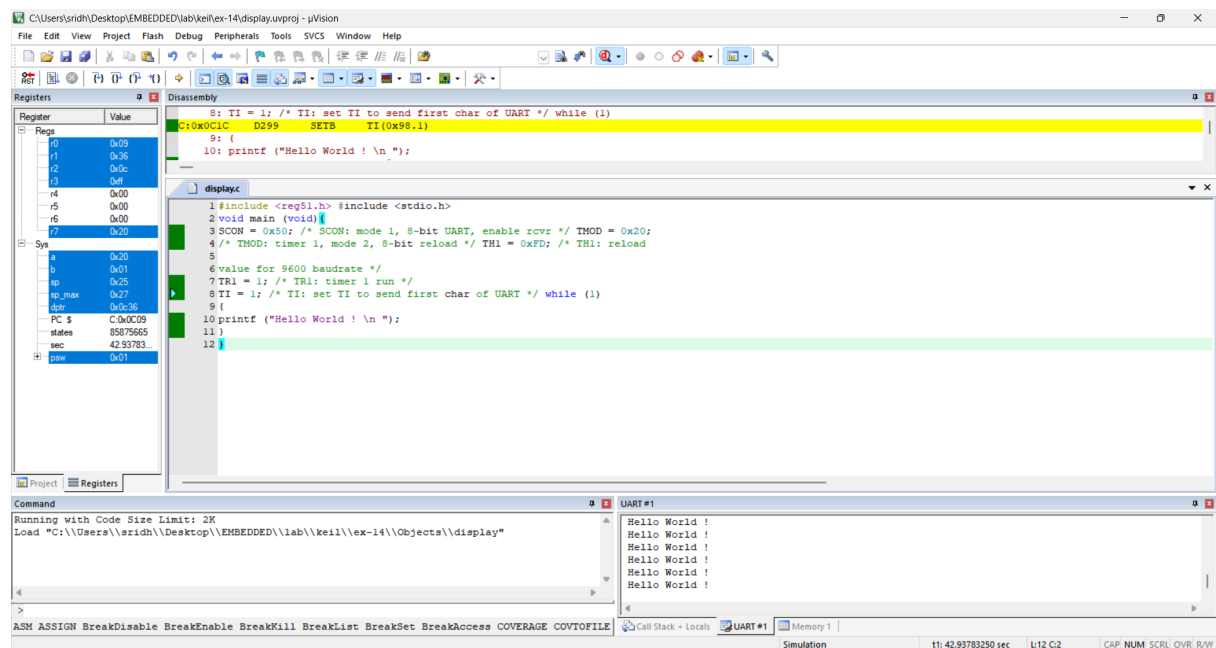
EXP 12: EMBEDDED C PROGRAM TO MULTIPLY TWO 16-BIT BINARY NUMBERS



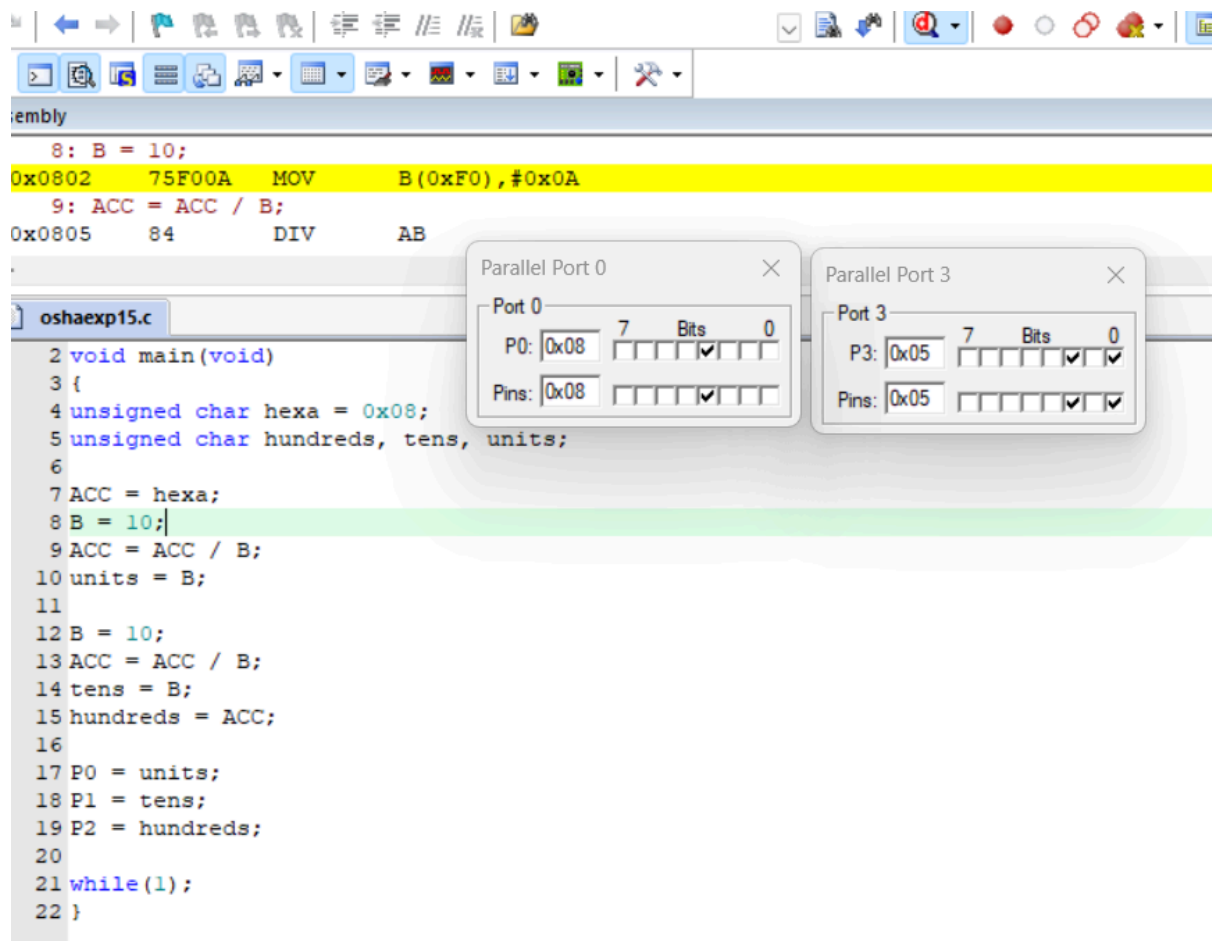
EXP 13: EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS AND STORE THE 32-BIT RESULT IN INTERNAL RAM



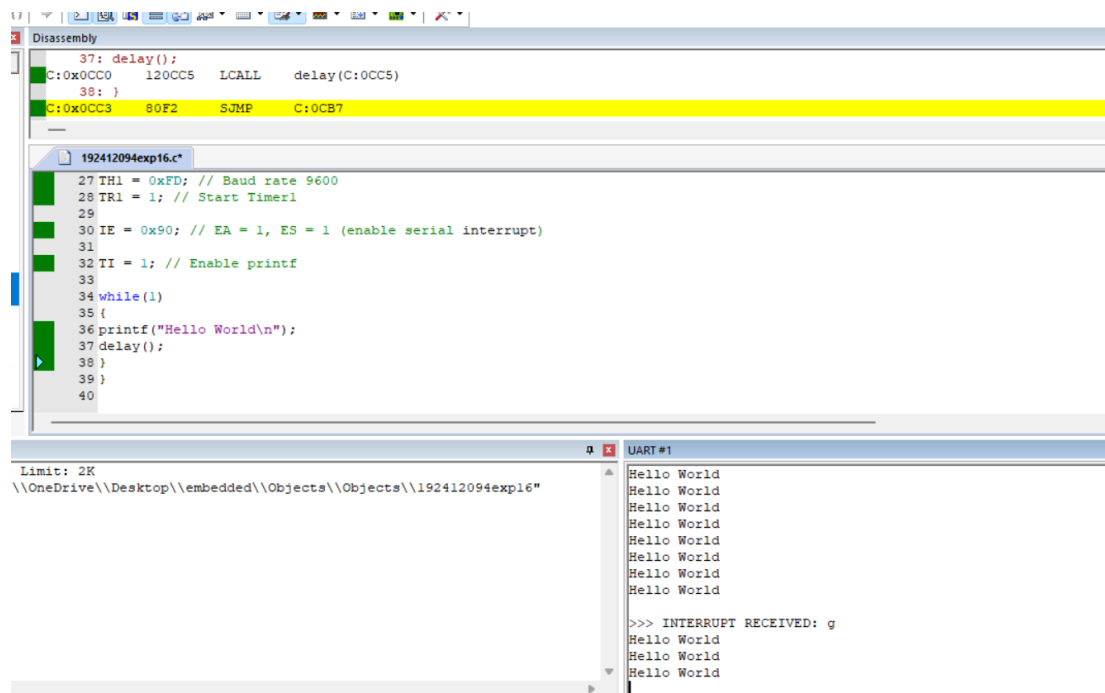
EXP 14: EMBEDDED C PROGRAM TO DISPLAY “HELLO WORLD” MESSAGE



EXP 15: EMBEDDED C PROGRAM TO CONVERT THE HEXADECIMAL TO DECIMAL AND DISPLAY THE DIGITS ON PORTS P0, P1 AND P2



EXP16 : SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL



EXP 17: UART INTERFACING USING INTERRUPT IN KEIL

The screenshot displays the Keil IDE environment for a project named '192412094EXP17.c'. The C code defines a timer interrupt service routine (ISR) and a main function. The ISR increments a counter and toggles a PWM signal. The main function configures the timer and enables the interrupt.

```

31: IE = 0x82; // EA = 1, ET0 = 1

7
8 void timer0_ISR(void) interrupt 1
9 {
10 TH0 = 0xFC; // Reload for 1ms delay
11 TL0 = 0x66;
12
13 count++;
14
15 if(count < duty)
16 PWM = 1; // ON time
17 else
18 PWM = 0; // OFF time
19
20 if(count >= 200) // Total period = 200ms
21
22 count = 0;
23 }
24
25 void main()
26 {
27 TH0 = 0xFC; // 1ms preload
28

```

Configuration windows shown:

- Parallel Port 1:** Port 1: 0x61, Bits: 0, Pins: 0x61.
- Interrupt System:**

Int Source	Vector	Mode	Req	Ena	Pri
P3.2/Int0	0003H	0	0	0	0
Timer 0	000BH	0	1	0	0
P3.3/Int1	0013H	0	0	0	0
Timer 1	001BH	0	0	0	0
Serial Rcv.	0023H	0	0	0	0
Serial Xmit.	002BH	0	0	0	0
- Serial Channel:** Mode: 8-Bit var Baudrate, SC0N: 0x54, SBUF: 0x00, REN: checked, RB8: checked, Baudrate: 0.

The 'Selected Interrupt' window shows 'EA' checked. The 'Location/Value' window shows 'C:0x08B8'.

EXP 18: Generation of pwm using timer interrupt

The screenshot shows the Keil IDE with a logic analyzer trace of a PWM signal on pin P1. The signal is a square wave with a period of 200ms and a duty cycle of approximately 66%. The C code defines the timer interrupt and the main function.

```

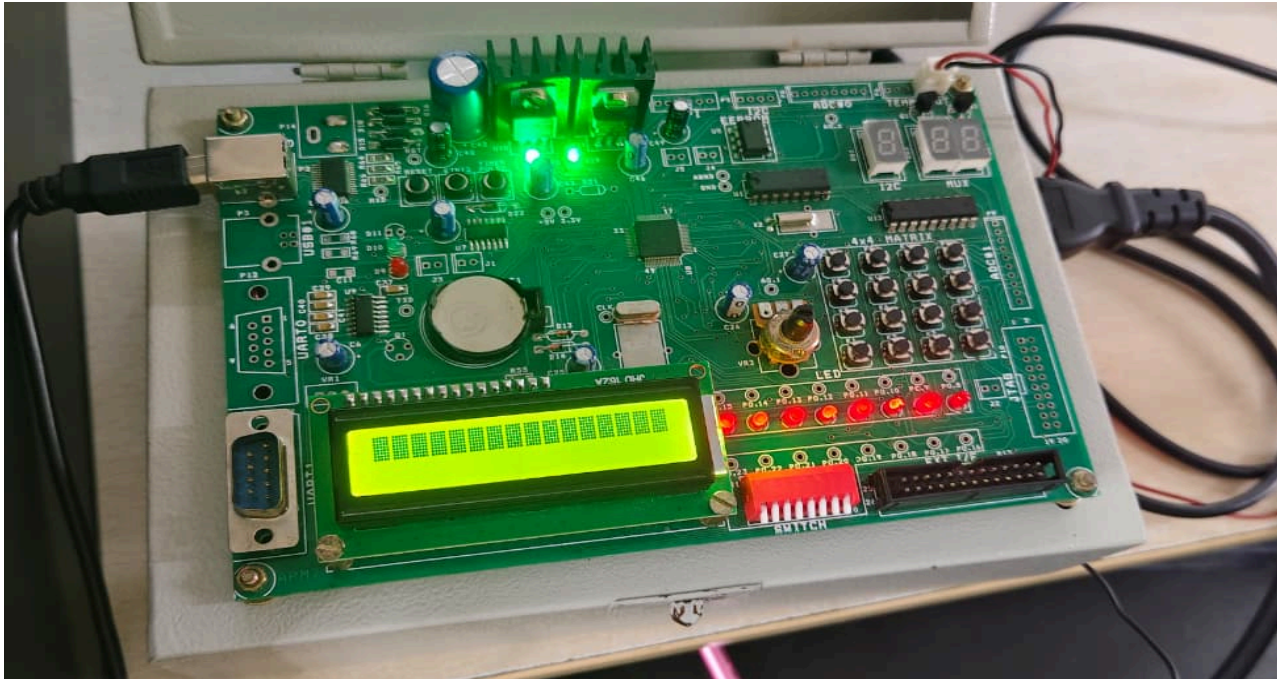
20 if(count >= 200) // Total period = 200ms
21
22 count = 0;
23
24 void main()
25 {
26 TH0 = 0xFC; // Timer0 Mode 1 (16-bit)
27 TH0 = 0xFC; // 1ms preload
28 TL0 = 0x66;
29
30 IE = 0x82; // EA = 1, ET0 = 1
31 TR0 = 1; // Start Timer0
32
33 while(1);
34 }

```

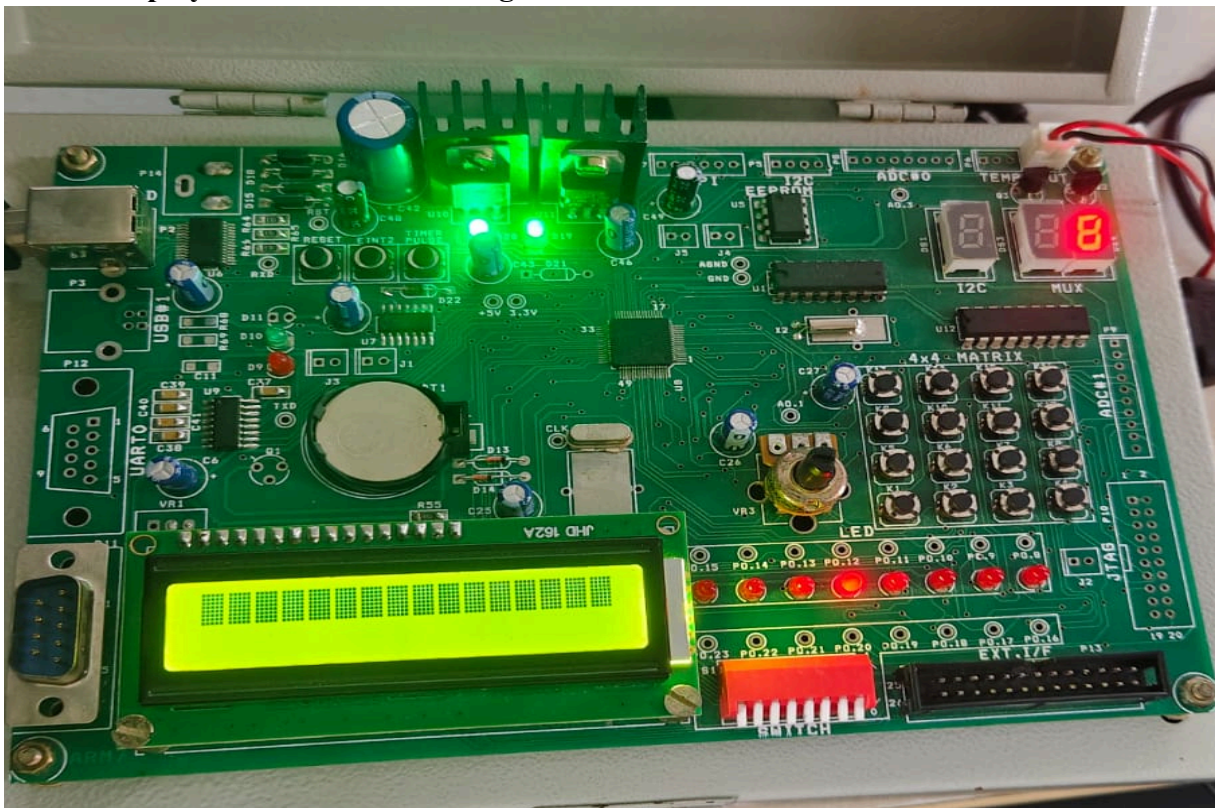
The logic analyzer trace shows the PWM signal on P1. The code window shows the main function and the timer interrupt service routine. The 'Command' window shows the command 'LA (P1 & 0x1) >> 0'. The 'Call Stack - Locals' window shows the 'MAIN' function at location 'C:0x08B8'.

EXP 19: Blinking LEDs using software delay routine in LPC2148 kit

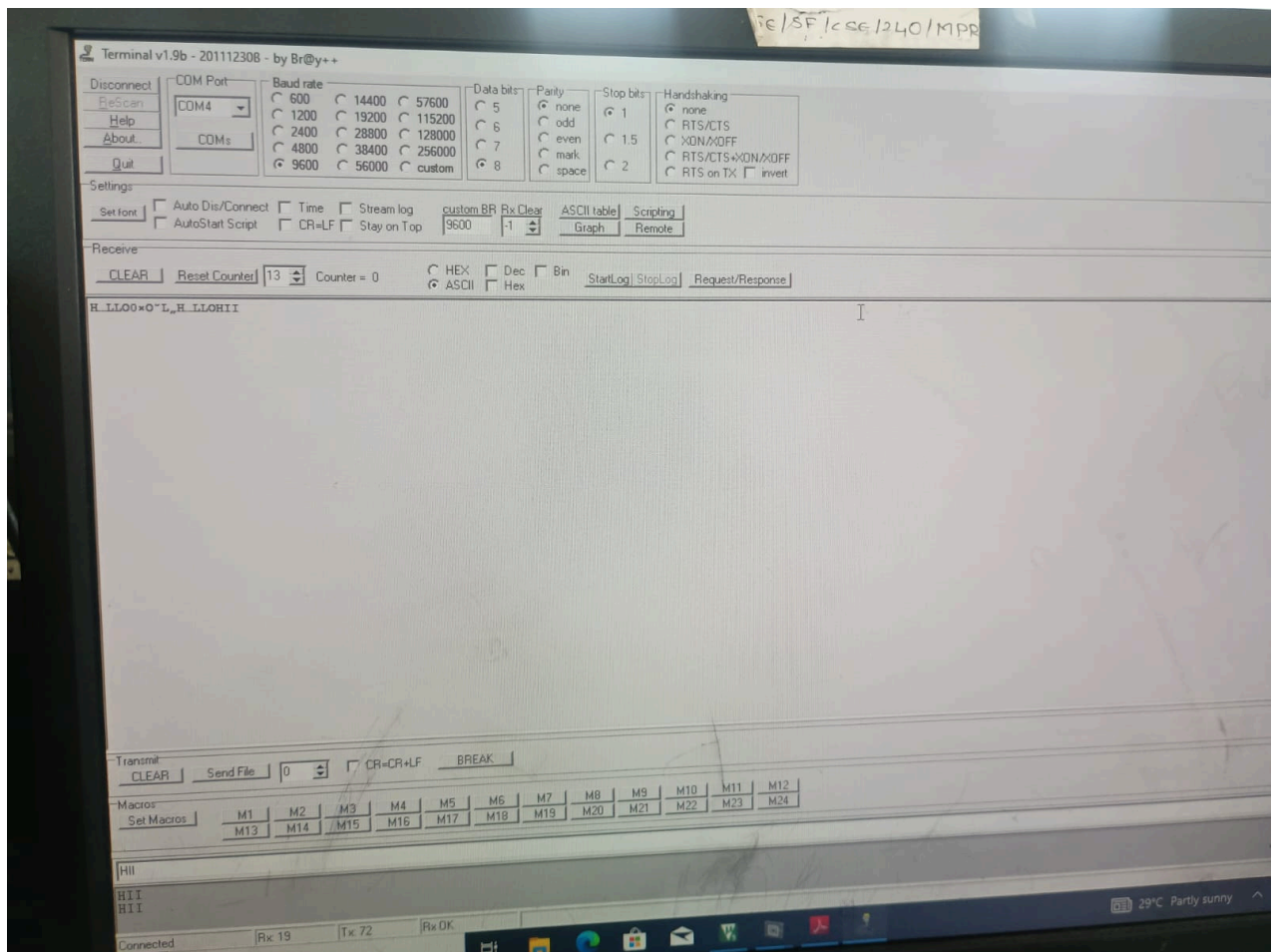
EX 20: Program to read the switch and display in the LEDs using LPC2148 kit



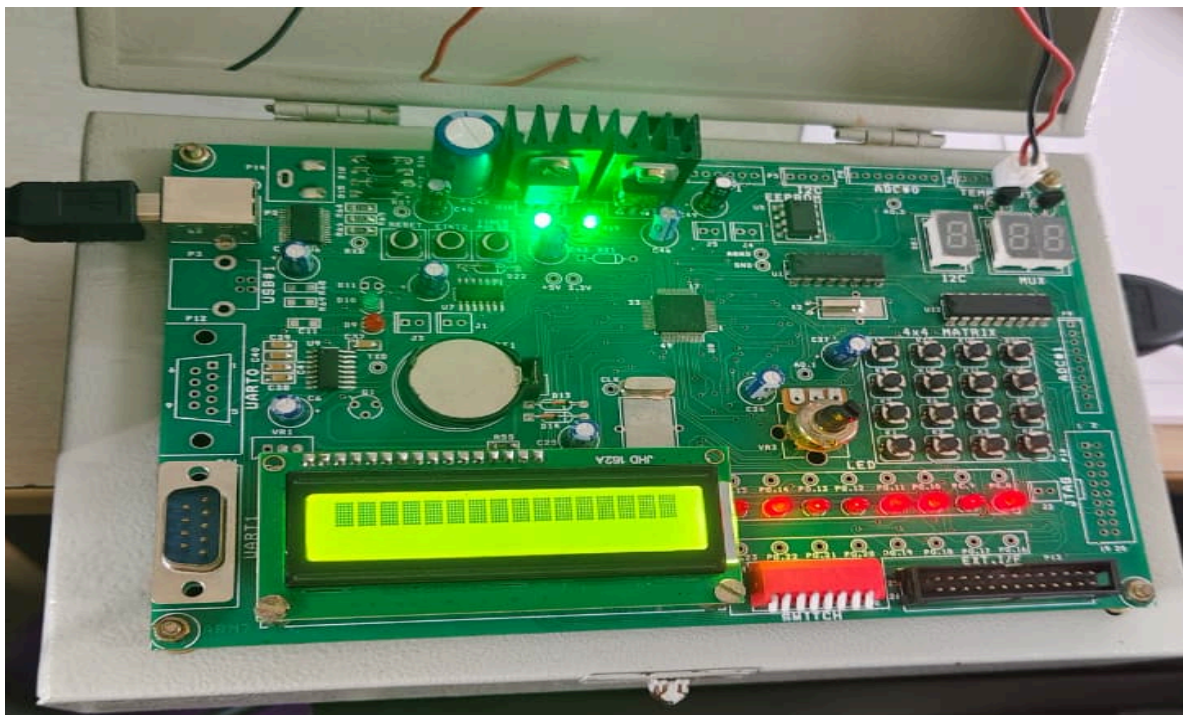
EX 21: Display a number in seven segment LED in LPC2148 kit



EX 22: Serial transmission and reception using on-chip UART in LPC2148 kit



EX 23: Access an internal ADC and display the binary output in LEDs in LPC2148 kit



Exp 24: Modeling Clock-Driven Concurrent Processes Using SC_THREAD and wait() in SystemC

c

```
saisri@DESKTOP-P73175QB:~/systemc_test/build$ ./test_systemc

SystemC 2.3.4-Accellera --- May 20 2025 11:42:17
Copyright (c) 1996-2022 by all Contributors,
ALL RIGHTS RESERVED
Time: 0 s - Clock Triggered
Time: 1 ns - Clock Triggered
Time: 2 ns - Clock Triggered
Time: 3 ns - Clock Triggered
Time: 4 ns - Clock Triggered
saisri@DESKTOP-P73175QB:~/systemc_test/build$
```

EXP 25: Producer-Consumer Problem Using SystemC

```
saisri@DESKTOP-P73175QB:~/systemc_test/build$ cd ~/systemc_test
saisri@DESKTOP-P73175QB:~/systemc_test$ cd build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ cmake ..
-- Configuring done (0.0s)
-- Generating done (0.0s)
-- Build files have been written to: /home/saisri/systemc_test/build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ make
[ 33%] Built target and_gate
[ 66%] Built target producer_consumer
[100%] Built target test_systemc
saisri@DESKTOP-P73175QB:~/systemc_test/build$ ./producer_consumer

SystemC 2.3.4-Accellera --- Aug 21 2026 05:34:04
Copyright (c) 1996-2022 by all Contributors,
ALL RIGHTS RESERVED
Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
saisri@DESKTOP-P73175QB:~/systemc_test/build$
```


EXP 26: Transaction Level Modeling (TLM) using SystemC

```
saisri@DESKTOP-P73175QB:~/systemc_test/build$ cd ..
saisri@DESKTOP-P73175QB:~/systemc_test$ ls
CMakeLists.txt  and_gate.cpp  build  producer_consumer.cpp  test_systemc.cpp  test_systemccc.cpp
saisri@DESKTOP-P73175QB:~/systemc_test$ nano tlm.cpp
saisri@DESKTOP-P73175QB:~/systemc_test$ nano CMakeLists.txt
saisri@DESKTOP-P73175QB:~/systemc_test$ cd build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ cmake ..
-- Configuring done (0.0s)
-- Generating done (0.0s)
-- Build files have been written to: /home/saisri/systemc_test/build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ make
[ 25%] Built target and_gate
[ 50%] Built target producer_consumer
[ 75%] Built target test_systemc
[ 87%] Building CXX object CMakeFiles/tlm.dir/tlm.cpp.o
[100%] Linking CXX executable tlm
[100%] Built target tlm
saisri@DESKTOP-P73175QB:~/systemc_test/build$ ./tlm
SystemC 2.3.4-Accellera --- Aug 21 2026 05:34:04
Copyright (c) 1996-2022 by all Contributors,
ALL RIGHTS RESERVED
Initiator sending: 555
Target received: 555
saisri@DESKTOP-P73175QB:~/systemc_test/build$
```

EXP 27: Binary Semaphore in SystemC

```
saisri@DESKTOP-P73175QB:~/systemc_test$ nano CMakeLists.txt
saisri@DESKTOP-P73175QB:~/systemc_test$ ls /opt/systemc
include  lib  share
saisri@DESKTOP-P73175QB:~/systemc_test$ ls /opt/systemc/lib
cmake  libsystemc.so  libsystemc.so.2.3  libsystemc.so.2.3.4
saisri@DESKTOP-P73175QB:~/systemc_test$ nano binary_semaphore.cpp
saisri@DESKTOP-P73175QB:~/systemc_test$ nano CMakeLists.txt
saisri@DESKTOP-P73175QB:~/systemc_test$ nano CMakeLists.txt
saisri@DESKTOP-P73175QB:~/systemc_test$ cd build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ cmake ..
-- Configuring done (0.1s)
-- Generating done (0.0s)
-- Build files have been written to: /home/saisri/systemc_test/build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ make
[ 20%] Built target and_gate
[ 40%] Built target producer_consumer
[ 60%] Built target test_systemc
[ 80%] Built target tlm
[ 90%] Building CXX object CMakeFiles/binary_semaphore.dir/binary_semaphore.cpp.o
[100%] Linking CXX executable binary_semaphore
[100%] Built target binary_semaphore
saisri@DESKTOP-P73175QB:~/systemc_test/build$ ./binary_semaphore
SystemC 2.3.4-Accellera --- Aug 21 2026 05:34:04
Copyright (c) 1996-2022 by all Contributors,
ALL RIGHTS RESERVED
Task1 Entered Critical Section at 1 s
Task1 Entered Critical Section at 4 s
Task1 Entered Critical Section at 7 s
saisri@DESKTOP-P73175QB:~/systemc_test/build$
```

```

[ 33%] Built target producer_consumer
[ 50%] Built target test_systemc
[ 66%] Built target tlm
[ 83%] Built target binary_semaphore
[ 91%] Building CXX object CMakeFiles/counting_semaphore.dir/counting_semaphore.cpp.o
[100%] Linking CXX executable counting_semaphore
[100%] Built target counting_semaphore
saisri@DESKTOP-P73175QB:~/systemc_test/build$ ./counting_semaphore
-bash: ./counting_semaphore: No such file or directory
saisri@DESKTOP-P73175QB:~/systemc_test/build$ ./counting_semaphore

SystemC 2.3.4-Accellera --- Aug 21 2026 05:34:04
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Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_0
In file: /home/saisri/systemc-2.3.4/src/sysc/kernel/sc_object_manager.cpp:153

Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_1
In file: /home/saisri/systemc-2.3.4/src/sysc/kernel/sc_object_manager.cpp:153
Task Accessing Resource at 1 s Remaining: 1
Task Accessing Resource at 1 s Remaining: 0
Task Accessing Resource at 3 s Remaining: 0
Task Accessing Resource at 4 s Remaining: 0
Task Accessing Resource at 5 s Remaining: 0
Task Accessing Resource at 6 s Remaining: 0
Task Accessing Resource at 7 s Remaining: 0
Task Accessing Resource at 8 s Remaining: 0
Task Accessing Resource at 9 s Remaining: 0
saisri@DESKTOP-P73175QB:~/systemc_test/build$ █

```

EXP 29: Priority-Based Interrupt Handling

```

Task Accessing Resource at 6 s Remaining: 0
Task Accessing Resource at 7 s Remaining: 0
Task Accessing Resource at 8 s Remaining: 0
Task Accessing Resource at 9 s Remaining: 0
saisri@DESKTOP-P73175QB:~/systemc_test/build$ cd ..
saisri@DESKTOP-P73175QB:~/systemc_test$ nano priority_interrupt.cpp
saisri@DESKTOP-P73175QB:~/systemc_test$ nano CMakeLists.txt
saisri@DESKTOP-P73175QB:~/systemc_test$ cd build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ cmake ..
-- Configuring done (0.0s)
-- Generating done (0.0s)
-- Build files have been written to: /home/saisri/systemc_test/build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ make
[ 14%] Built target and_gate
[ 28%] Built target producer_consumer
[ 42%] Built target test_systemc
[ 57%] Built target tlm
[ 71%] Built target binary_semaphore
[ 85%] Built target counting_semaphore
[ 92%] Building CXX object CMakeFiles/priority_interrupt.dir/priority_interrupt.cpp.o
[100%] Linking CXX executable priority_interrupt
[100%] Built target priority_interrupt
saisri@DESKTOP-P73175QB:~/systemc_test/build$ ./priority_interrupt

SystemC 2.3.4-Accellera --- Aug 21 2026 05:34:04
Copyright (c) 1996-2022 by all Contributors,
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Low Priority Interrupt at 1 s
High Priority Interrupt at 2 s
saisri@DESKTOP-P73175QB:~/systemc_test/build$ █

```

```
-- Build files have been written to: /home/saisri/systemc_test/build
saisri@DESKTOP-P73175QB:~/systemc_test/build$ make
[ 12%] Built target and_gate
[ 25%] Built target producer_consumer
[ 37%] Built target test_systemc
[ 50%] Built target tlm
[ 62%] Built target binary_semaphore
[ 75%] Built target counting_semaphore
[ 87%] Built target priority_interrupt
[ 93%] Building CXX object CMakeFiles/rtos_scheduler.dir/rtos_scheduler.cpp.o
[100%] Linking CXX executable rtos_scheduler
[100%] Built target rtos_scheduler
saisri@DESKTOP-P73175QB:~/systemc_test/build$ ./rtos_scheduler

SystemC 2.3.4-Accellera --- Aug 21 2026 05:34:04
Copyright (c) 1996-2022 by all Contributors,
ALL RIGHTS RESERVED
Task1 Running at 0 s
Task2 Running at 0 s
Task1 Running at 1 s
Task2 Running at 1 s
Task1 Running at 2 s
Task2 Running at 2 s
Task1 Running at 3 s
Task2 Running at 3 s
Task1 Running at 4 s
Task2 Running at 4 s
Task1 Running at 5 s
Task2 Running at 5 s
saisri@DESKTOP-P73175QB:~/systemc_test/build$ |
```